Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

4. Let $G = (V, E)$ be a weighted graph with $V = \{0, 1, 2, 3, 4, 5\}$ and the edge between i and j with weight w represented by the triple (i, j, w) in $E = \{(0, 5, 5), (1, 3, 8), (1, 2, 6), (2, 5, 5), (0, 2, 1), (3, 5, 4), (4, 5, 7), (3, 4, 5), (1, 4, 5)\}$.
   (a) Construct the minimum spanning tree with Prim's algorithm. Show all steps.
   (b) When implementing Prim's algorithm, explain how the choice of the data structure leads to an $O(m \log(n))$ asymptotic cost for graphs of n vertices and m edges.

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Answer:

4. (a)
   i. $S = \{0\}$, $T = \emptyset$
   ii. $S = \{0, 2\}$, $T = \{(0, 2)\}$
   iii. $S = \{0, 2, 5\}$, $T = \{(0, 2), (0, 5)\}$
   iv. $S = \{0, 2, 5, 3\}$, $T = \{(0, 2), (0, 5), (3, 5)\}$
   v. $S = \{0, 2, 5, 3, 1\}$, $T = \{(0, 2), (0, 5), (3, 5), (1, 2)\}$
   vi. $S = \{0, 2, 5, 3, 1, 4\}$, $T = \{(0, 2), (0, 5), (3, 5), (1, 2), (1, 4)\}$

   (b) Prim's algorithm uses a heap to store the vertices, where the key for each vertex v in the heap is the minimum cost to attach v to the set S of already connected vertices.

   The operations to manipulate a heap of size n, where n is the number of vertices, have a cost of $O(\log(n))$.

   Changing the attachment cost happens at most once per edge, and for m edges we thus arrive at the $O(m \log(n))$ cost for Prim's algorithm.